

NeuroDrive Prototype Operational Workflow

How Physical Motion Transforms into Real-Time Browser Gameplay

1. Executive Operational Pipeline

1

Physical Motion Sensing (MPU6050)

The 6-axis IMU reads angular velocity (gyroscope) and acceleration forces (accelerometer) at 20 Hz.

2

I2C Transmission & On-Chip Smoothing (ESP32)

ESP32 reads 14-byte raw burst data over I2C (GPIO 19/18) and applies Exponential Moving Average filtering.

3

Wi-Fi AP & Wireless JSON Telemetry Host

ESP32 broadcasts softAP `MPU6050\_Controller` and serves filtered JSON via `/data` REST endpoint.

4

Client UI & High-Frequency Polling

Connected browser (phone/PC) fetches `/data` every 50ms and computes deadzone-calibrated steering & speed inputs.

5

Real-Time Web Engine Execution

JavaScript game loop updates car dynamics, obstacle spawning, hitboxes, score, and Web Audio SFX.

2. Hardware & Firmware Data Flow

The prototype functions completely standalone without needing external internet or third-party routers. The ESP32 micro-controller serves as both the hardware interface and the web host.

HARDWARE LAYER

- **Sensor Burst Read:** Starts at I2C address **0x3B** to pull 14 continuous bytes.
- **Pinout Mapping:** SDA on GPIO 19, SCL on GPIO 18.
- **Baud Rate:** Serial debugging output at 115,200 baud.
- **Wi-Fi Hotspot:** SoftAP running on SSID `MPU6050_Controller`.

SIGNAL PROCESSING LAYER

To prevent game jitter caused by handheld tremors, the firmware applies a low-pass Exponential Moving Average (EMA) filter:

$$y_t = 0.35 \cdot x_t + 0.65 \cdot y_{t-1}$$

- **Gyro X (*gx*):** Filtered for steering sensitivity.
- **Accel Y (*ay*):** Filtered for acceleration / braking pitch.

3. Client-Side Browser Mechanics

INPUT PROCESSING

- **Zero Calibration:** A 1.2-second sample window records resting baseline orientation before gameplay starts.
- **Deadzone Handling:** Ignores minor gyro fluctuations (± 400 units) to prevent unwanted drifting.

GAME MECHANICS & AUDIO

- **Speed Control:** Tilt forward to accelerate (up to ~22 units/s); tilt back to brake (min 3 units/s).
- **Entities & Collisions:** Coins (+25 pts x combo), Shields, Rocks, Cones, and Rival cars.

- **Steering Proportionality:** Car target position lerps smoothly based on roll tilt angle.
- **Web Audio API:** Synthesizes real-time sound effects (coin chime, crash rumble, pass beep) via oscillator nodes.

4. Component Interaction Summary

Component	Primary Role	Key Technology	Latency / Refresh Rate
MPU6050 IMU	Captures physical 6-DOF motion data	I2C Protocol	Hardware Sampling Rate
ESP32 Microcontroller	Signal filtering, HTTP Server & Access Point	Arduino C++ / WebServer library	Continuous Loop Execution
JSON Telemetry API	Wireless data bridge between MCU & Browser	HTTP GET (/data)	50 ms Polling Interval (20 Hz)
Web Browser Engine	Renders UI, calculates physics & plays sfx	JS / HTML5 DOM / Web Audio API	60 FPS (requestAnimationFrame)